

Colour Sensor Adhesive Inspection System

Document Type	Standard Operating Procedure (SOP)
Intended Audience	Department Head / Technical Supervisor / Operator
System	Automated Adhesive Inspection System — ESP32 Based
Revision	1.1

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1. System Overview

The Colour Sensor Adhesive Inspection System is an automated quality inspection station designed to verify the adhesive application on two components — the **Ring** and the **Hub** — before allowing the machine to proceed to the next production step.

The system rotates both parts through a full 360° inspection using a stepper motor-driven fixture. Two dedicated colour sensors continuously monitor the adhesive application during rotation. At the end of each inspection cycle, the system automatically determines whether both parts pass or fail and signals the result to the operator through dedicated indicators and buzzers.

The system is designed to eliminate manual adhesive inspection errors and ensure only correctly applied parts proceed in the production process.

2. Panel & Hardware Layout

The system consists of a table-mounted control panel with two sides:

Operator Side

Accessible to the production operator at all times.

Component	Description
Power Switch	Panel-mounted ON/OFF switch — controls main power to the entire system
Cycle Start Button	Momentary push button — starts the inspection cycle
Ring Reject Buzzer	Audible buzzer — sounds when Ring part fails inspection
Hub Reject Buzzer	Audible buzzer — sounds when Hub part fails inspection
Machine OK Indicator	Green indicator light — illuminates when both parts pass inspection

Supervisor Side

Accessible to the supervisor or department head only.

Component	Description
Bypass Key Switch	Key-operated switch — allows supervisor to reset a rejected cycle and force machine enable

Internal Components (Not Operator Accessible)

Component	Description
SMPS Power Supply	Converts mains power to DC — powers the control system and colour sensors
ESP32 Controller	Main control unit — runs the inspection logic
PCF8574 I/O Expanders	Handles all digital inputs and outputs
Stepper Motor Driver	Drives the fixture rotation motor

3. Indicators & Buzzers Reference

Quick Reference Table

Signal	Meaning	Action Required
Green Indicator ON	Inspection passed — machine enabled	No action — machine proceeds automatically
Green Indicator SOLID (no cycle running)	Bypass mode is active	Supervisor awareness required
Ring Buzzer ON	Ring part adhesive inspection FAILED	Supervisor must use bypass key to reset
Hub Buzzer ON	Hub part adhesive inspection FAILED	Supervisor must use bypass key to reset
Both Buzzers ON	Both Ring and Hub parts FAILED	Supervisor must use bypass key to reset
All OFF	System idle — waiting for parts and cycle start	Load parts and press Cycle Start

4. System Startup Procedure

Follow these steps every time the system is powered on:

Step 1 — Pre-power Check

- Ensure the fixture is empty — no parts loaded
- Ensure the bypass key switch is in the OFF position
- Ensure no objects are obstructing the fixture rotation area

Step 2 — Power ON

- Turn the panel power switch to the ON position
- The SMPS will power up and supply DC power to the controller and sensors

Step 3 — System Boot

- The ESP32 controller will initialize automatically
- Allow approximately 3 seconds for the system to complete startup
- All indicators and buzzers will remain OFF after boot — this is normal

Step 4 — System Ready

- The system is now in IDLE state and ready to accept the first inspection cycle
- No indicators will be ON in idle state

Note: If the bypass key switch is accidentally left ON during power-up, the system will enter bypass mode automatically. Turn the key switch OFF to return to normal operation.

5. Normal Cycle Operation

Follow this procedure for every inspection cycle:

Step 1 — Load Parts

- Place the Ring part in its designated fixture position
- Place the Hub part in its designated fixture position
- Ensure both parts are correctly seated — incorrect placement may cause false readings

Step 2 — Start Inspection

- Press the Cycle Start button on the operator panel
- The fixture will begin rotating immediately

Step 3 — Inspection Running

- The fixture completes one full 360° rotation
- Both colour sensors continuously monitor the Ring and Hub during rotation
- Do not touch the fixture or parts during rotation
- Do not press any buttons during the cycle

Step 4 — Read Result

- When rotation is complete the system will display the result automatically
- See Section 6 for result interpretation

Step 5 — After OK Result

- The Green Indicator will illuminate for 3 seconds to signal the result
- After 3 seconds, the system waits for the operator to remove both parts from the fixture
- Once both parts are removed and confirmed stable for 5 seconds, the Machine Relay enables automatically
- The Green Indicator stays ON for 30 seconds while the machine is enabled
- After 30 seconds the relay turns OFF and the system returns to IDLE — ready for next cycle

Step 6 — After Reject Result

See Section 7 — Reject Procedure.

6. Understanding Inspection Results

Result: PASS (OK)

Signal	What It Means
Green Indicator ON	Both Ring and Hub adhesive inspection passed
Machine relay enables after part removal	System has confirmed parts removed and machine is enabled

What to do: Remove both parts from the fixture. Wait for green indicator to confirm machine enable. Proceed with production.

Result: REJECT

Signal	What It Means
Ring Buzzer sounding	Ring part adhesive inspection failed
Hub Buzzer sounding	Hub part adhesive inspection failed
Both Buzzers sounding	Both Ring and Hub adhesive inspection failed

What to do: Do not remove parts immediately. Call the supervisor. See Section 7.

7. Reject Procedure

When a reject is detected the system latches the reject state and sounds the appropriate buzzer(s). The system will not allow a new cycle to start until the supervisor resets it using the bypass key.

Step-by-Step Reject Procedure

Step 1 — Identify the Reject

- Note which buzzer is sounding — Ring, Hub, or both
- This identifies which part has failed adhesive inspection

Step 2 — Inform Supervisor

- Operator must inform the supervisor immediately
- Do not attempt to restart the system without supervisor authorisation

Step 3 — Supervisor Inspection

- Supervisor visually inspects the rejected part(s)

- Supervisor decides whether to: remove and scrap the rejected part(s) and load new parts, or use bypass mode if production must continue (see Section 8)

Step 4 — Supervisor Resets Using Bypass Key

- 1 Supervisor turns the bypass key switch to the ON position
- 2 Buzzers will turn OFF
- 3 System will enter bypass mode — relay turns ON, green indicator turns ON solid
- 4 Supervisor turns the bypass key switch back to the OFF position
- 5 System returns to IDLE state — green indicator turns OFF

Step 5 — Load New Parts and Continue

- Operator loads new parts into the fixture
- Operator presses Cycle Start to begin the next inspection

Important: The reject latch is a deliberate safety feature. Only the supervisor bypass key can clear a reject. This ensures every reject is reviewed by a responsible person before production continues.

8. Bypass Mode — Supervisor Only

Bypass mode is a supervisor-controlled feature that forces the machine relay ON regardless of inspection results. It is intended for use in specific production situations authorised by the department head.

When to Use Bypass Mode

- When a known-good batch needs to proceed without inspection (e.g. during sensor maintenance)
- When instructed by the department head or quality team
- To reset the system after a reject event (see Section 7)

How to Activate Bypass Mode

- 1 Turn the bypass key switch to the ON position
- 2 The system will enter bypass mode: machine relay turns ON, green indicator turns ON solid, all buzzers turn OFF
- 3 The system will remain in bypass mode as long as the key switch is ON

How to Deactivate Bypass Mode

- 1 Turn the bypass key switch to the OFF position
- 2 The system will return to IDLE state
- 3 All outputs turn OFF
- 4 Normal inspection operation resumes

Warning: In bypass mode the inspection system does not check adhesive quality. The machine relay is forced ON regardless of part condition. Bypass mode should only be used with full authorisation and awareness of quality implications.

9. Part Changeover Procedure — Fixture Change & Sensor Teaching

Use this procedure whenever the system needs to switch to inspecting a **different Ring or Hub part variant**. This procedure must be completed in full before running production — an incomplete or incorrect sensor teach will cause the system to reject good parts or, worse, pass bad parts, because all inspection decisions depend entirely on correctly taught sensors.

This procedure has two parts: **(A) Fixture Change** (mechanical) and **(B) Sensor Teaching** (per sensor). Both must be completed for every part changeover.

A. Fixture Change

- 1 Loosen the fixture screw located underneath the fixture, using an Allen key
- 2 Lift the old fixture off the shaft
- 3 Place the new fixture (matching the new part variant) onto the shaft
- 4 Tighten the fixture screw securely

B. Sensor Teaching

Sensor teaching must be carried out **separately for the Ring sensor and the Hub sensor**. Complete the full teaching procedure for one sensor before starting the other. Each sensor has its own **SET** button and status LED located on the sensor body.

Before starting: After power-on, check that both sensors' LEDs are lit and indicating normally. If a sensor shows no LED activity, do not proceed — see Section 11, Troubleshooting.

Teach OK (Good) Part

- 1 Place a known-good part in front of the sensor being taught
- 2 Press the SET button on that sensor once
- 3 Confirm the sensor LED turns ON — this confirms the sensor has saved the OK reference to memory

Teach NG (Reject) Part

- 1 Remove the OK part and place a known-bad (NG) part in front of the same sensor
- 2 Press the SET button on that sensor once
- 3 Confirm the sensor LED turns OFF as soon as the NG part is detected — this confirms the sensor correctly distinguishes the NG condition

Complete the Changeover

- 1 Repeat the OK and NG teaching steps above for the second sensor
- 2 Once both sensors are taught, remove the teaching parts, load a real part, and run one test cycle to confirm correct OK/REJECT response before resuming production

Important: Do not skip the NG teach step. A sensor taught only with an OK part has no reference for what a bad part looks like and may accept incorrect parts as OK.

10. System Shutdown Procedure

Step 1 — Complete Current Cycle

- Ensure the current inspection cycle is complete before shutdown
- Do not power off during an active cycle

Step 2 — Remove Parts

- Remove all parts from the fixture

Step 3 — Confirm Idle

- Confirm all indicators and buzzers are OFF
- Confirm the bypass key switch is in the OFF position

Step 4 — Power OFF

- Turn the panel power switch to the OFF position
- The system will shut down immediately

11. Troubleshooting Guide

System Does Not Power On

Possible Cause	Action
Mains supply not connected	Check power cable and wall socket
Panel power switch faulty	Check switch continuity — replace if needed
SMPS failure	Check SMPS output voltage — replace if no output

System Powers On but No Response to Cycle Start Button

Possible Cause	Action
System not fully booted	Wait 5 seconds after power on and try again
Bypass mode active	Check bypass key switch — turn to OFF if ON
System in reject state	Supervisor must reset using bypass key (see Section 7)
Cycle Start button wiring fault	Check button wiring and connection to control panel

Fixture Does Not Rotate After Cycle Start

Possible Cause	Action
Motor driver not powered	Check stepper driver power supply
Motor cable disconnected	Check motor cable connections
Motor enable signal fault	Contact system technician

Green Indicator Does Not Come ON After Good Cycle

Possible Cause	Action
Parts not fully removed from fixture	Remove both parts completely and wait 5 seconds
Sensor detecting parts as still present	Check sensor alignment and cleanliness
Indicator lamp failure	Check indicator lamp — replace if needed

Ring Buzzer Sounds on Every Cycle Even with Correct Parts

Possible Cause	Action
Ring colour sensor dirty or misaligned	Clean sensor lens — check mounting position
Ring part not seated correctly in fixture	Ensure Ring is placed in correct position
Sensor wiring fault	Contact system technician

Hub Buzzer Sounds on Every Cycle Even with Correct Parts

Possible Cause	Action
Hub colour sensor dirty or misaligned	Clean sensor lens — check mounting position
Hub part not seated correctly in fixture	Ensure Hub is placed in correct position
Sensor wiring fault	Contact system technician

Both Buzzers Sound on Every Cycle

Possible Cause	Action
Both sensors dirty or misaligned	Clean both sensor lenses
Wrong parts loaded	Verify correct Ring and Hub parts are being used
System sensor calibration required	Contact system technician for recalibration

System Rejects Every Part After a Changeover

Possible Cause	Action
Sensor(s) not taught, or taught incorrectly	Repeat the Sensor Teaching procedure (Section 9) for the affected sensor

Machine Relay Does Not Enable After Part Removal

Possible Cause	Action
Parts not fully removed — sensors still triggered	Remove parts completely from fixture
Removal not stable for 5 seconds	Keep parts away from fixture for at least 5 seconds
Relay output fault	Contact system technician

Bypass Mode Will Not Deactivate

Possible Cause	Action
Key switch faulty	Check key switch wiring
Key switch not fully turned to OFF	Ensure key switch is fully in OFF position

12. Safety Notes

- Do not reach into the fixture rotation area during an active cycle
- Do not bypass the inspection system without supervisor authorisation
- Do not attempt to repair or modify the control panel without isolating mains power first
- Do not operate the system if any wiring is visibly damaged
- Do not clean sensors with liquids while the system is powered
- The bypass key switch must remain in the supervisor's possession at all times
- Any persistent faults not resolved by this guide must be reported to the system technician immediately
- In case of emergency, use the panel power switch to shut down the entire system immediately

*Document prepared by system integrator. For technical support, contact the system supplier.
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